

Flow rate : flow units / time unit

Flow time : time through process

Little's Law : Average Inventory = avg flow rate * avg flow time

Flow time : Inventory / Flow rate

Inventory turns = 1 / Flow time

Inventory turns (financial) = Average inventory / COGS

Per-unit inventory costs (%) = Annual inventory cost (%) / Annual inventory terms

Process capacity = Minimum { capacity of resource 1, ..., capacity of resource n }

Process flow rate = Minimum { Average input, Demand, Process capacity }

Utilization (%) = Flow rate / capacity

Implied utilization (%) = Demand / capacity

Capacity = Resources / Processing time

Time through empty process

worker-paced, = Sum { resource 1 processing time, ..., resource n processing time }

machine paced = Number of resources in sequence * processing time at bottleneck

Time to finish X units

starting with empty process = Time through empty process + (X – 1) / Flow rate

with a continuous-flow process = Time through empty process + (X / Flow rate)

Labor content = Sum of processing times with labor, often as time per unit

Cost of direct labor = Total wages per unit of time / Flow rate per same unit of time

= wages per week / units produced per week

= \$/unit

Average labor utilization (%) = Labor content * Flow rate / Number of workers

Cycle time : time between units of output = 1 / Flow rate

Idle time across all workers at resource I =

Cycle Time * (Number of workers at resource i) – Processing time at resource I

Cost of direct labor (\$/unit) = Total wages per unit of time / Flow rate per unit of time

Average labor utilization (%) = Labor content / (Labor content + total idle time)

Capacity given batch size (units/time) = Batch size / (Setup time + Batch size * Processing time)

Average inventory (units) = $\frac{1}{2}$ * Batch size * (1 – Flow rate * Processing time)

Recommended batch size = (Flow rate * Setup time) / (1 – Flow rate * Processing time)

Average inventory = Order quantity / 2

Annual inventory costs (\$/unit or % of unit cost):

accounts for costs of capital, costs of storage, costs of obsolescence

Inventory costs (\$/unit of time) for order quantity Q and inventory cost h

= $\frac{1}{2}$ * Q * h

Order setup costs (\$/unit of time) for setup cost K, flow rate R. and order quantity Q

= K / (Q / R)

= (K * R) / Q

Economic order Quantity (Q*) = sqrt ((2 * Setup cost * Flow rate) / Holding cost)

Cost per unit under EOQ with setup cost K, holding cost h, and flow rate R

= sqrt ((2 * K * h) / R)

Source: Cachon, G., & Terwiesch, C. (2024). Matching supply with demand: An introduction to operations management. McGraw Hill. Compiled by Zach Rose, licensed under CC0 1.0 Universal: <https://creativecommons.org/publicdomain/zero/1.0/>

